

Error: The difference of magnitude between measured value (A_m) and true value (A_t) is called Error.

- In Engineering, Error is represented with $\pm$
- In general, Error means (Absolute error) i.e.,
 1. No plus, No minus
 2. Only magnitude
- Error represented by ' ϵ '

$$\epsilon = A_m - A_t$$

($\therefore A_m$ is measured value, A_t is true value)

These errors are broadly classified into two types:

1. Static error (error is constant w.r.t time)
2. Dynamic error (error is changing w.r.t time)

Static error: The error which is independent of time is called Static error.

Dynamic error : The error which depends on time is called dynamic error.

+ve $\Rightarrow A_m > A_t$

-ve $\Rightarrow A_m < A_t$

Error may be +ve (or) -ve

Correction Factor (CF) :

The value which we are added (or) subtracted from the measured value in order to get the true value (or) actual value is called Correction factor.

It may be +ve (or) -ve

C.F = -(ϵ)

For ex: Error (ϵ) = ($A_m - A_t$) = 0.2

Correction factor = -(ϵ) = -0.2

So the measured value decreases by '0.2'

Relative Static Error (R.S.E):

The static error taken over the true value is called relative static error.

$$R.S.E = \frac{E}{A_t}$$

$$R.S.E = \frac{A_m - A_t}{A_t}$$

$$\%R.S.E = \frac{A_m - A_t}{A_t} \times 100$$

Basic characteristics of an Instrument:

1. Accuracy
2. Precision
3. Repeatability
4. Reproducibility
5. Sensitivity
6. Resolution
7. Hysteresis
8. Dead time and Dead zone
9. Drift

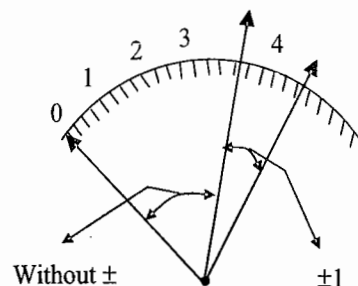
1. Accuracy: It is the degree of closeness of measured quantity to true quantity. (i.e., conformity to truth)

2. %GAE (Guaranteed Accuracy Error) :

When accuracy is mentioned in terms of error it's called "Guaranteed Accuracy Error"

If this error expressed w.r.t full scale deflection it's called %GAE.

It is expressed by Manufacturer, it is constant for meter.



Error is represented by ' $\pm$ '

Deflection is represented without ' $\pm$ '



Instrument	%GAE	
A	$\pm 1\%$	Error
B	$\pm 99\%$	
C	98%	Deflection
D	1%	

So while choosing meter error should be as minimum as possible & Deflection should be as high as possible.

3. Limiting error (or) Guaranteed error:

It's calculated from %GAE

L.E is constant for a meter and it's also expressed by manufacturer

L.E give the range of output for a given input reading.

% Limiting error: If the error is expressed w.r.t true value is called %L.E

$$\% \text{ Limiting error} = \frac{A_m - A_t}{A_t} \times 100$$

It's a variable error and it is calculated by observer.

[In solving problems, %GAE = Accuracy]

$$\% \text{ Accuracy} = (1 - |\epsilon_r|) \times 100$$

Note: For a good instrument %GAE, limiting error and %L.E must be minimum and accuracy (or) %accuracy must be high.

Ex: $\% \epsilon_r = \pm 2\% \Rightarrow \text{Accuracy} = 98\%$

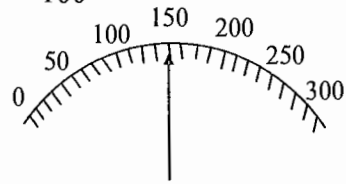
When $\epsilon_r = 0.02$ (when measuring 1V with 100 V full scale) then $\% \text{ Accuracy} = (1 - |\epsilon_r|) \times 100$
 $= (1 - 0.02) \times 100 = 98\%$

Example 1:

If (0 – 300 V) voltmeter has an accuracy of $\pm 1\%$, then find %LE, when measuring

- (a) 200 V (b) 150 V
(c) 100 V (d) 50 V

Sol: $L.E = \frac{1}{100} \times 300 = 3V$



(a) $\%LE \text{ of } 200 \text{ V} = \pm \frac{3}{200} \times 100 = \pm 1.5\%$

(b) $\%LE \text{ of } 150 \text{ V} = \pm \frac{3}{150} \times 100 = \pm 2\%$

(c) $\%LE \text{ of } 100 \text{ V} = \pm \frac{3}{100} \times 100 = \pm 3\%$

(d) $\%LE \text{ of } 50 \text{ V} = \pm \frac{3}{50} \times 100 = \pm 6\%$

From the above result we can conclude that :
 As the value to be measured increased, then %L.E decreases.

All indicating instruments give more %LE below half of f.s.d of scale.

Example 2:

To measure 10Volts, if one selects a (0–100)volts range voltmeter whose accuracy is within $\pm 1\%$, then the error in this measurement is _____

- (a) $\pm 7.5\%$ (b) $\pm 5\%$
(c) $\pm 10\%$ (d) $\pm 20\%$

Ans: (c)

Sol: (0–100)V

$$\% \text{ G.A.E} = \pm 1\%$$

$$100 \times \frac{\pm 1}{100} = \pm 1$$

$$\text{Error} = 100 \times \frac{\pm 1}{10} = \pm 10\%$$

Example 3:

Three DC currents I_1 , I_2 and I_3 meet at a node with I_1 entering and I_2 and I_3 leaving the node. I_1 and I_2 are measured as 100mA and 99mA with a $\pm 1\%$ accuracy. Then the value of I_3 and the accuracy of I_3 are.

- (a) $1A \pm 2$ (b) $199mA \pm 2$
(c) $1mA \pm 2$ (d) $1mA \pm 1.99$



Ans: (d)

Sol: $I_1 = 100\text{mA}$

$I_2 = 99\text{mA}$

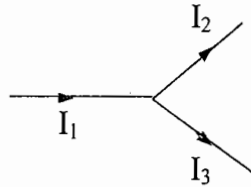
Accuracy = 1%

By KCL

$I_1 + (-I_2) = I_3$

$100\text{mA} \pm 1\% - 99\text{mA} \pm 1\% = I_3$

$I_3 = 1\text{mA} \pm 1.99$



Example 4:

An ammeter has a range of 0 to 30A. The Instrument gave the following readings.

Current flow (A) 0 5 10 15 20 25

Ammeter reading (A) 1 4 12 14 22 28

The non-linearity of the instrument in terms of full scale reading

(a) 20% (b) 10% (c) 25% (d) 15%

Ans: (b)

Sol: Full scale reading = 30V

$E_{\text{error}} = 28 - 25 = 3\text{V}$

$\therefore 3\text{V}$ is a 10% of full scale Reading

Precision:

It indicates the degree of closeness between the measured values.

For a good instrument, high precision is preferred when the measured values are very close.

Ex: While measuring '10A', precision instrument measures 9.9, 9.8, 9.7, 9.6

- An high precision instrument may be accurate (or) may not be accurate i.e., precision doesn't give the guarantee about accuracy.
- But an high accuracy instrument is a precision instrument (i.e.,) accuracy guarantees the precision

Note:

For a good instrument, accuracy is at most important condition but precision is of second order preferences.

Repeatability:

It indicates the degree of closeness among measure values, when same input value

measurand is measured number of times by same instrument with observer under same conditions.

Reproducibility :

It also indicates the degree of closeness, when the same input value is measured number of times by one instrument with many number of observers, under different conditions.

- Precision is the measurement of repeatability (or) reproducibility.
- Precision also depends on number of significant figures in digital instruments.
- As the number of significant figures increases then actual information conveyed also increases i.e. precision increases.

Ex: 130 (3 significant) = here chance of lead (129 or 131)

If 130.00 V (5 significant figures) = (129.9 or 130.1) Here precision increases.

Example 5:

Two Resistors R_1 and R_2 are connected in series with $R_1 = 28.7\Omega$ and $R_2 = 3.624\Omega$.

Calculate the total resistance to the appropriate number of significant figures.

- (a) 32.32 (b) 32.324
(c) 32.3 (d) 32.3240

Ans: (c)

Sol: $R_1 = 28.7\Omega$ (Three significant figure)

$R_2 = 3.624\Omega$ (Four significant figure)

Total Resistance $R_1 = 28.7 + 3.624$

$= 32.324\Omega$

(Five significant figures)

$= 32.3$

(Three significant figures)

This is done because one of the resistances is accurate only to three significant figures and therefore the result should also be reduced to three significant figures and hence the value 32.3Ω

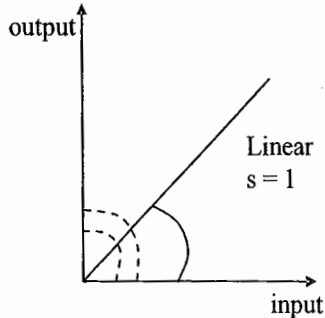
Sensitivity (or) Figure of merit (or) Scale factor

(S): It indicates the relation between change in output readings for change in input reading for an instrument.

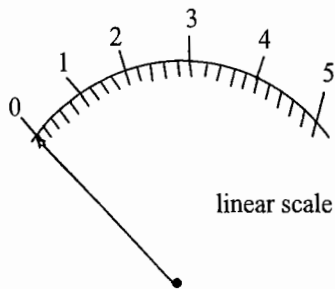


This relation may be linear (or) Non-linear.

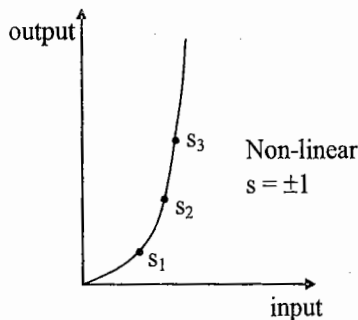
$$s = \frac{\text{change in output}}{\text{change in input}} = \tan \theta$$



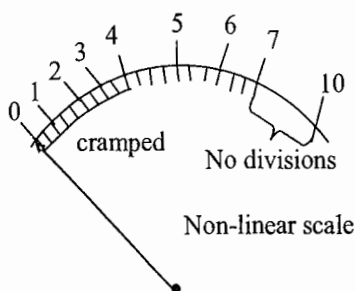
$$\Rightarrow \theta \propto I$$



Ex: PMMC, Dynamometer



$$\Rightarrow \theta \propto I^2$$



Ex: MI

- For an instrument $s = 1$ preferred. It indicates the change in output is proportional to change in input.
- $s < 1$ (or) $s > 1$ indicates non-linear reading, then the scale is cramped at the beginning and almost no divisions at the end.

Note: Sensitivity playing an important role in designing of type of scale at the time of designing

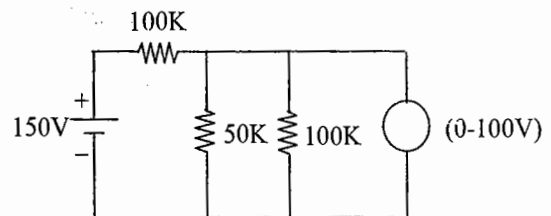
It's the reciprocal of full scale deflection current.

In order to reduce the loading error, high sensitive instruments are preferred.

Example 6:

Two voltmeters with the ranges of (0 – 100 V) and has the sensitivities of 1000 Ω/V , 2000 Ω/V . If these voltmeters are connected individually to measure the voltages across 50 K Ω resistors then find the corresponding readings and % error.

Sol:



$$V_{50K\Omega} = 150 \times \frac{50}{150} = 50V$$

(i) V_A

$R_A = 1000 \times 100 = 100 \text{ k}\Omega$ (resistance of voltmeter which is across it)

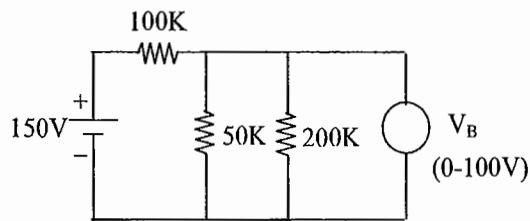
$$50 \text{ K} \parallel 100 \text{ K} = 33.3 \text{ k}\Omega$$

$$V_A = V_{33.3K\Omega} = 150 \times \frac{33.3k\Omega}{100K + 33.3K}$$

$$\% \text{error} = \frac{37.5 - 50}{50} \times 100 = -25\%$$



(ii)



$$R_B = 20000 \times 100 = 200 \text{ K}\Omega$$

$$50 \text{ K}\Omega \parallel 200 \text{ K}\Omega = 40 \text{ K}\Omega$$

$$V_B = V_{48.78} = \frac{150 \times 40}{(40 + 100)} = 42.85 \text{ V}$$

$$\% \text{error} = \frac{42.85 - 50}{50} \times 100 = -14.3\%$$

Loading effect:

When a voltmeter is connected across a resistor to measure voltage, voltmeter draws current for its working, then voltmeter internal resistance and the resistance of voltage to be measured, resultant resistance decreases, then voltmeter reads less than the actual value. It's called loading effect.

Note :

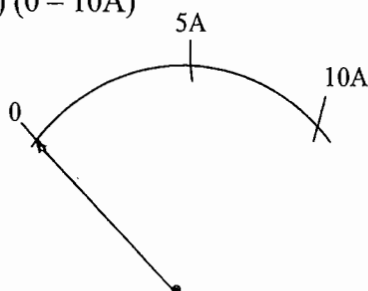
To reduce loading effect, high sensitivity voltmeters to be selected for the same range.

- Compared to electrical instruments, electronic instruments have high sensitivity and more accurate.

Resolution (or) Discrimination

Even a smallest change in input, for which there is a definite numerical change in output, this numerical change is called resolution.

(a) (0 – 10A)



$$\text{Resolution} = \frac{10}{2} = 5 \text{ A / division}$$

$$\text{Resolution} = \frac{\text{full scale deflection (fsd)}}{\text{No. of divisions}}$$

- So as number of divisions increases, resolution increases for the same range.

Note: For non-linear scale we can't calculate resolution.

Hysteresis Error:

If the meter gives less than the actual value for ascending values and shows more than the actual values for descending values.

If the meter gives two different readings for same input value is called hysteresis error.

Causes:

1. Electric Hysteresis:

The meter has two different flux densities while increasing and decreasing time of the same input value.

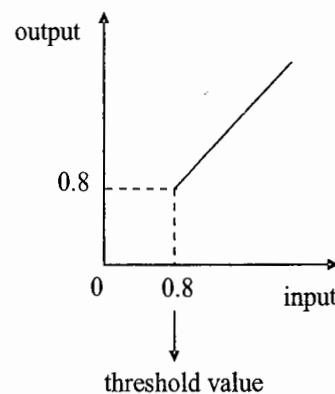
2. Mechanical Hysteresis:

Whatever the stress is developed while loading the meter, the same stress can't be removed.

Threshold:

If the input to the instruments is increased gradually from zero value, then a minimum value is required on a input to get a detachable change in output.

The minimum value is called threshold.





Dead time:

The pointer takes some minimum time to start from its zero position, because of inertia of moving system. The minimum time is called Dead time.

Dead zone: For a large change in input for which there is no output for some portion of the scale is called Dead zone.

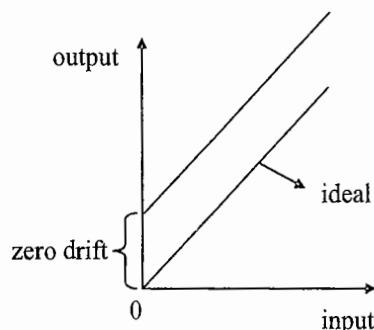
Drift:

It's an error; but constant throughout the range of scale.

Zero drift:

For an instrument, there is variation in output, which is not caused by change in input.

- It's due to instability of components and internal temperature change.



- For a good instrument, accuracy, precision, repeatability, reproducibility, sensitivity, resolution, linearity, fidelity (dynamic error is zero) are desirable characteristics.
- For a good instrument, limiting error, hysteresis error, dead time, dead zone, drift and non-linearity are undesirable characteristics.

Example 7:

The following readings are taken by one voltmeter to the same input values are 102.1 V, 102.4V, 102.3 V, 102.6 V, 102.7 V, 102.9V, then find the range of errors.

Sol: $V_{avg} = \frac{102.1 + 102.4 + 102.3 + 102.6 + 102.7 + 102.9}{6}$

$= 102.5 \text{ V} \rightarrow \text{True value } (A_t)$

From question

$V_{min} = 102.1$

$V_{max} = 102.9$

$V_{max} - V_{avg} = 102.9 - 102.5 = 0.4$

$V_{avg} - V_{min} = 102.5 - 102.1 = 0.4$

Range of error $= \pm \frac{(V_{avg} - V_{min}) + (V_{max} - V_{avg})}{2}$

$= \pm \frac{0.4 + 0.4}{2} = \pm 0.4$

Note: To obtain true value, measure the input value number of times and take the average. If it is infinite times, 100% true value is obtained

Types of Static Errors

- Gross errors:** These are completely due to human mistakes i.e., doing loose connections, reading the values, taking the values, memorizing and doing calculations. Gross errors are not common for all observers. These are variable errors.

Note: The reading can be taken correctly from the meter when meter is in vertical position and eye contact should be over the pointer (i.e., 90° to the meter)

- Systematic error:**
 - Instrumental
 - Environmental
 - Observational

(i) Instrumental :

- Due to inherent short coming of instrument, ageing effect, frictional error, Hysteresis error.
- Misuse of an instrument : choosing wrong range of an instrument.



(c) loading effect

(ii) Environmental error:

Due to temperature, pressure, humidity, dust etc.

(iii) Observational error: or (parallax error)

Note: To eliminate the shadow of the pointer a mirror strip is placed in between scale and pointer (parallel error is eliminated)

Systematic errors are constant errors these are common for all observers.

3. Random errors or accidental errors:

These are due to the un identified sources These errors comes suddenly and eliminate suddenly but reason is not known.

These are more but effect is very small to eliminate the random error statistical analysis is carried out i.e. multi sample data in which the same input value is measured no. of times which are instrument by one observer or many observers.

$x_1, x_1, x_1, \dots, x_n \Rightarrow 'n' \text{ times}$

Average value $(\bar{x}) = \frac{x_1 + x_2 + \dots + x_n}{n}$

2. Deviation $(d) \Rightarrow d_1 = x_1 - \bar{x}$

$$d_2 = x_2 - \bar{x}$$

$$\therefore d_n = x_n - \bar{x}$$

3. Average deviation $= \frac{|d_1| + |d_2| + \dots + |d_n|}{n}$

To know precision of an instrument.

4. Standard deviation: Deviation from average value

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + \dots + d_n^2}{n}} \quad (n > 20)$$

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2}{n-1}} \quad (n \leq 20)$$

5. Probable error (r) = $\pm 0.6745\sigma$

6. Probable error of mean = $(r_m) = \frac{r}{\sqrt{n-1}}$

7. Uncertainty Error

$$X = f(x_1, x_2, x_3, \dots, x_n)$$

$W_{x1}, W_{x2}, \dots, W_{xn}$ are the uncertainties of

$x_1, x_2, \dots, x_n$

Uncertainty of $x = w_x =$

$$\pm \sqrt{\left(\frac{\partial x}{\partial x_1}\right)^2 \times w_{x1}^2 + \left(\frac{\partial x}{\partial x_2}\right)^2 \times w_{x2}^2 + \dots + \left(\frac{\partial x}{\partial x_n}\right)^2 \times w_{xn}^2}$$

8. Standard Error

$$X = f(x_1, x_2, x_3, \dots, x_n)$$

$\sigma_{x1}, \sigma_{x2}, \dots, \sigma_{xn}$ are the standard deviation error of $x_1, x_2, \dots, x_n$

Standard deviation error of ' x ' = $\sigma_x =$

$$\pm \sqrt{\left(\frac{\partial x}{\partial x_1}\right)^2 \times \sigma_{x1}^2 + \left(\frac{\partial x}{\partial x_2}\right)^2 \times \sigma_{x2}^2 + \dots + \left(\frac{\partial x}{\partial x_n}\right)^2 \times \sigma_{xn}^2}$$

Note:

For the calculation of uncertainty and standard deviations convert percentage values into real values and substitute in formulae.

Example 8:

Uncertainty distribution is used for

(a) analysis of multi-sample data

(b) analysis of single sample data

(c) analysis of both single and multiple sample data

(d) none of the above

Ans: (b)



Example 9:

A set of independent 5 measurements were made to determine the weight of a lead shot. The weights in gramme were

1.570, 1.597, 1.591, 1.562, 1.577

Find the probable error of the mean

- (a) 4.9×10^{-3} (b) 11.6×10^{-3}
(c) 14.5×10^{-3} (d) 1.57

Ans: (a)

Sol: Find arithmetic mean

$$= \frac{1.570 + 1.597 + 1.591 + 1.562 + 1.577}{5}$$

= 1.5794 gramme

Deviation from the mean

$$d_1 = 1.570 - 1.5794 = -0.0094$$

$$d_2 = 1.597 - 1.5794 = 0.0176$$

$$d_3 = 1.591 - 1.5794 = 0.0116$$

$$d_4 = 1.562 - 1.5794 = -0.0174$$

$$d_5 = 1.577 - 1.5794 = -0.0024$$

Stranded deviation

$$\sigma = \sqrt{\frac{d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2}{5-1}} = 0.0145.$$

$$\text{Problem error of the mean} = \frac{0.6745 \sigma}{\sqrt{5-1}} \\ = 4.9 \text{m gram}$$

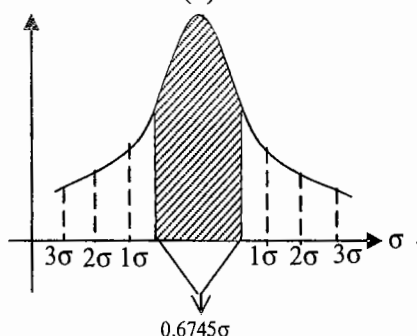
Example 10:

In which of following deviation's has the highest probability range in specifying the measured data.

- (a) $\pm 3\sigma$ limit (b) $\pm 2\sigma$ limit
(c) Probable error (d) Standard deviation

Ans: (a)

Sol:



From graph we can approximated as
Deviation Probability

$$0.6745 \sigma \rightarrow \approx 50\% \text{ (Probable error)}$$

$$\pm \sigma \rightarrow \approx 68.28\% \text{ (Standard deviation)}$$

$$\pm 2\sigma \rightarrow \approx 95.46\%$$

$$\pm 3\sigma \rightarrow \approx 99.74\%$$

Example 11:

The manufactures providing four meter with labeling their accuracies given below

(1) Meter "A" with Accuracy $\pm 1\%$

(2) Meter "B" with Accuracy $\pm 98\%$

(3) Meter "C" with Accuracy 99%

(4) Meter "D" with Accuracy 1%

Among the given above meters, the high accurate meters are

- (a) only BD (b) only BC
(c) only AC (d) only AD

Ans: (c)

Sol: For example let us consider we are measuring 5A current by using above given meters then

Meter A gives $\Rightarrow 5 \pm 1\% \Rightarrow 5 \pm 0.01 \Rightarrow 4.99$
(or) 5.01

Meter B gives $\Rightarrow 5 \pm 98\% \Rightarrow 5 \pm 0.98 \Rightarrow 4.02$
(or) 5.98

Meter C gives $\Rightarrow 99\% \text{ of } 5A \Rightarrow 4.95A$

Meter D gives $\Rightarrow 1\% \text{ of } 5A \Rightarrow 0.05A$

Conclusion: from above we can concluded meter "A" and meter C gives the highest accuracy readings among the given meters.

Example 12:

Power factor of a circuit is determined by :

$$\cos \phi = \frac{P}{VI}$$

Where P = Power in watt

V = Voltage in volt

I = current in Ampere

The relative error in power, current and voltage are



respectively $\pm 0.5\%$; $\pm 1\%$; $\pm 1\%$ calculate the relative error in power factor. Also calculate the uncertainty in the power factor if the errors are specified as uncertainties

- (a) 2.5%;1.8 (b) 2.3%; 1.5
(c) 2.5%;1.89 (d) 2.5%; 1.5

Ans: (d)

Sol: Given data

Relative error in power $\Rightarrow \pm 0.5\%$

Relative error in voltage $\Rightarrow \pm 1\%$

Relative error in current $\Rightarrow \pm 1\%$

$$\text{Given that } \cos\phi = \frac{P}{VI}$$

$$\begin{aligned}\therefore \text{Relative error in power factor} \\ &= 0.5\% + 1\% + 1\% \\ &= 2.5\%\end{aligned}$$